



MASTER'S THESIS

Multimodal Representation Learning and Interaction-Aware Object Segmentation for Behaviour Understanding

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Abstract

Understanding human and animal behaviour from video requires models that can exploit heterogeneous sensor streams and accurately localise manipulated objects at the pixel level. This report details contributions made during a research internship at the Mathis Group (EPFL) from May 2026 through November 2026, documenting research milestones achieved through September 2026 across two complementary tracks.

The first contribution, STrOMAE (Shared Transformer Orchestrator Masked Autoencoder), is a unified, modality-agnostic fine-tuning framework built upon VideoMAE. Prior laboratory workflows maintained fragmented, modality-specific codebases across video, audio, and segmentation maps. STrOMAE unifies these into a single architecture where each modality is mapped to a token sequence by a dedicated embedder, concatenated, and jointly processed by a shared Vision Transformer trunk with arbitrary multi-task prediction heads. The pipeline is declaratively configured, eliminating custom per-dataset training code. We validate the framework by reproducing baseline benchmarks on EPFL-Smart-Kitchen-30 and MammAlps.

The second contribution addresses interaction-aware object segmentation on the EPFL-Smart-Kitchen-30 dataset, motivated by the downstream goal of 3D object and scene reconstruction to analyze gaze–action dynamics in stroke rehabilitation. Given nine calibrated multi-view video streams, 3D hand poses, and dense action annotations, we formulate an orchestration framework to extract persistent 2D masks across viewpoints. We systematically benchmark foundation segmentation models (SAM 3) across distinct manipulation regimes, demonstrating that failure modes are predominantly view- and illumination-dependent rather than occlusion-driven. Building on these findings, we propose a geometry-first pipeline that leverages multi-camera calibration and action priors to robustly track and segment manipulated objects without requiring dense ground-truth training masks.

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1 Introduction

1.1 Context and Motivation

This internship is conducted within the Mathis Group for Computational Neuroscience & AI at EPFL from May 1 to November 1, 2026, under the supervision of Alexander Mathis and Andy Bonnetto. This report documents the contributions achieved through the September 2026 milestone. The laboratory operates at the interface of computer vision, machine learning, and computational neuroscience, developing modern artificial intelligence methods to model, measure, and understand complex natural behaviour.

First, the group builds and releases *densely instrumented* multimodal datasets. The EPFL-Smart-Kitchen-30 dataset [3] records 16 subjects cooking four recipes over 29.7 hours, observed simultaneously by nine calibrated static RGB-D cameras, one head-mounted HoloLens 2 (providing egocentric video, eye gaze, and hand tracking), and inertial measurement units. Action sequences are annotated at a density of roughly 34 segments per minute, with SMPL body meshes [23] and MANO hand meshes [29] fitted at every frame in a shared global coordinate frame. This project is expanding as standardized Smart Kitchen environments are deployed across collaborating European research laboratories. The MammAlps dataset [11] provides a wildlife counterpart: nine camera traps in the Swiss National Park recording synchronised video, audio, and reference segmentation maps, densely labelled for species and hierarchical behaviour.

Second, because these datasets are multimodal and multi-view, modeling workflows face two recurring challenges: scaling a single representation model across arbitrary input modalities, and extracting reliable pixel-level object representations from multi-camera rigs. My internship addressed one of each.

1.2 Problem Formulations

Project A : STrOMAE: Shared Transformer Orchestrator MAE. Masked autoencoders [16] and their video extension VideoMAE [34] have become the default pre-training recipe for video representation learning. In practice, however, each experiment in the group had its own fork: a video-only experiment had one training script and one loader, a video+audio experiment had another, and a video+audio+segmentation experiment had a third. The three forks diverged, shared no evaluation code, and made cross-experiment comparison unreliable. The question I was asked to answer was: *can a single architecture and a single training loop serve any combination of modalities and any set of prediction targets, without per-dataset code?*

Project B : Interaction-Aware Object Segmentation. EPFL-Smart-Kitchen-30 annotates *what* action occurs and *when*, and collects the 3D pose of the hands and body, but it does not localise the manipulated objects in the image. The long term goal of the EPFL-Smart-Kitchen-30 project is to have a better understanding of kinetics in stroke patients. In order to do so we think correlation between eye gaze and objects positions could be informative. Recovering that information, for each frame, the pixel mask of the object currently being manipulated, would close the loop between the kinematic annotations and the visual scene. The question here was: *given nine calibrated views, dense action annotations, and 3D hand pose, how well can we segment the manipulated object, and what is the bottleneck?*

Both questions were answered in the affirmative, though not in the form initially expected. For the first, I designed and implemented STrOMAE, a token-level multimodal architecture in which an entire experiment, its modalities, its prediction heads, its loss weights, is specified declaratively, so that adding a dataset requires no new training code. Getting it to train correctly required a controlled over-fitting protocol that surfaced several defects in the inherited codebase, none of which raised an error. For the second, the answer came from a benchmark rather than a model: measuring where SAM 3 actually fails on EPFL-Smart-Kitchen-30 footage

showed that hand-object occlusion is not the dominant failure mode, which redirected the project from learning an interaction prior towards orchestrating a frozen segmenter across the nine calibrated views.

2 Background and Related Work

2.1 Behaviour recognition and multimodal representation learning

The framework developed here inherits its backbone from the masked autoencoder line of work. Masked autoencoding [16] showed that reconstructing heavily masked image patches with an asymmetric encoder-decoder yields transferable ViT [9] representations, and VideoMAE [34] carried the recipe to video by masking spatiotemporal tubelets at extreme ratios, a design paralleled by spatiotemporal masked autoencoding [10]. This project does not repeat that pretraining stage. It initialises from the publicly released VideoMAE ViT-B and ViT-L checkpoints distributed with InternVideo [36] and treats masked pretraining as a fixed upstream asset, so the contribution sits entirely downstream of the pretrained trunk. Patch embedding, tubelet size and layer-wise learning rate decay follow the VideoMAE fine-tuning protocol, as does the feature extractor of TIM [5], from which the training loop is adapted.

Extending masked representations across modalities has been approached in two broad ways. AudioMAE [17] applies the same objective to spectrograms, while CAV-MAE [13] and attention bottleneck fusion [25] keep one encoder per modality and introduce a narrow late fusion interface between them. MultiMAE [1] instead trains a single encoder over concatenated tokens drawn from RGB, depth and semantic maps. The architecture here follows that second strategy but relocates it to the fine-tuning stage: each modality is projected by its own lightweight embedder into a common token space, receives its own learned positional table, and the resulting tokens are concatenated into one sequence consumed by a single shared transformer trunk. Only the video embedder inherits pretrained weights; the audio, segmentation-map, pose and auxiliary-view embedders begin from random initialisation and acquire meaning only during fine-tuning, which is the principal asymmetry this design accepts in exchange for not requiring multimodal pretraining. Because modalities and output heads are declared in a configuration file and resolved through a registry, adding a stream or a label space requires no change to the trunk or to the data loader. Supervision is multi-head in the EPIC-KITCHENS sense [7], with one loss per head and configurable per-head weights.

MammAlps [11] supplies the second evaluation setting and contrasts sharply with EPFL-Smart-Kitchen-30. It records wild mammals through camera traps in the Swiss National Park, so the imagery is nocturnal, low-resolution and unposed, the subjects are animals rather than instructed participants, and the audio track carries ecologically meaningful signal rather than incidental noise. Where EPFL-Smart-Kitchen-30 offers dense, calibrated, human-centred capture, MammAlps offers scarce, uncontrolled, species-varied capture, which makes the pair a useful test of whether one shared trunk transfers across acquisition regimes instead of overfitting a single rig.

The MammAlps benchmark used here asks for clip-level prediction of three label spaces simultaneously: a multi-label set of fine-grained actions, a single-label activity, and the species present, scored by mean average precision over classes, on the division shipped with the Zenodo release. The EPFL-Smart-Kitchen-30 behaviour recognition benchmark [3] defines clip-level joint verb and noun classification over the dense action segments introduced in Section 2.6, with 31 verb and 78 noun classes, evaluated by top-1 accuracy per head under a subject-disjoint split. Several deviations from the published protocols must be stated before the results of Section 3.3 can be read as comparisons. The modality combinations reported here are chosen to exercise the shared trunk and do not necessarily match those of either reference baseline, so a difference in score may reflect a difference in inputs rather than in method. The ViT-L initialisation used

for the stronger runs is Kinetics-700 fine-tuned and is therefore a more favourable starting point than a plain masked-pretrained backbone. Balanced sampling and per-head loss weighting are training-side choices made here and are not part of either published recipe, and the learning rate is scaled by effective batch size, so raw rates are not directly comparable across tables. Finally, the MammAlps figures reported here concern the first release and its clip-level benchmark; the later season-two release defines a different task, per-frame detection with temporal retrieval, and no comparison against its published numbers is claimed.

2.2 Promptable segmentation: SAM, SAM 2, SAM 3

The Segment Anything Model [18] reframed segmentation as a promptable task: given an image and a geometric prompt (a point, a box), return the mask of the indicated object. SAM 2 [26] extended this to video with a memory bank that propagates a mask across frames, producing a temporally consistent object track known as a masklet.

SAM 3 [4] is the model this project builds on. Its new capability, promptable concept segmentation (PCS), takes a short noun phrase (or an image exemplar) and returns both masks and identities for all matching instances in an image or video. Architecturally it couples an image-level detector with a memory-based video tracker sharing one backbone, and decouples recognition from localisation with a dedicated presence head. It is reported to roughly double the accuracy of previous systems on image and video PCS.

The practical significance for this project is that SAM 3 collapses into a single text-driven model what used to be a three-model pipeline: an open-vocabulary detector [20, 21, 38], a promptable segmenter, and a tracker, assembled by systems such as Grounded-SAM [27] and DEVA [6]. Starting from a frozen SAM 3 is therefore the sensible baseline, and the research question becomes what additional signal the EPFL-Smart-Kitchen-30 rig can inject that SAM 3 has no access to.

Two properties of PCS matter for what follows. First, the prompt is a concept, so the returned set is every instance matching that concept, and selecting which instance is the manipulated one is left entirely to the user of the model. Second, a concept is defined by its text, so an object whose appearance changes enough to leave the extension of that text ceases to be detected at all. Both are structural rather than incidental, and both are addressed in Section 4.3.1.

2.3 Interactive hand-object segmentation

The task of segmenting not just objects but specifically the objects a person is interacting with was formalised by EgoHOS [37], which released 11,243 egocentric images with per-pixel labels for left hand, right hand, left-hand object, right-hand object and two-hand object, and was the first to label hand-object contact boundaries.

CaRe-Ego [31], is directly relevant to the design I initially proposed. It contributes two ideas. The Hand-guided Object Feature Enhancer (HOFE) uses hand features as queries in a cross-attention block over object features, so that the object representation is conditioned on the hand rather than merely concatenated with it. The Contact-centric Object Decoupling Strategy (CODS) disentangles the overlapping label attributes of two-hand objects during training, which removes an ambiguity that otherwise diverts capacity away from contact modelling. I thought the idea of creating an enriched hand-object embedding in order for the detector to be aware of such interactions could be a lead into our goal.

Crucially, CaRe-Ego operates on single egocentric images, with no 3D hand pose, no multi-view information, and no foundation-model backbone. The signals that EPFL-Smart-Kitchen-30 provides, namely fitted MANO meshes in a shared world frame, dense action annotations and nine calibrated viewpoints, are precisely the ones this line of work has had to do without.

2.4 Object transformation and state change

Cooking is dominated by actions that do not merely move an object but change what it is: a pepper is halved, then diced; mushrooms are sliced into a pile. For a concept-driven segmenter this is not a tracking difficulty but a recognition one, since the transformed object may no longer fall under the noun phrase that retrieved it.

VOST [33] isolates exactly this failure mode, benchmarking video object segmentation under transformations such as cutting, breaking and folding, and shows that methods relying on appearance consistency degrade sharply once an object’s shape and topology change. Complementary work targets the recognition side rather than the mask: ChangeIt [30] learns object states and state-modifying actions from unlabelled video, and Ego4D [14] includes an object state change classification benchmark.

SPOC [24] is the closest methodological neighbour to the approach taken here. It introduces spatially-progressing state change segmentation, which labels which regions of an object have already been transformed, and notably does not segment anything itself: it consumes mask proposals from a frozen Grounding DINO + SAM + tracker stack and trains a small transformer to classify each proposal as actionable, transformed, or background, supervised by CLIP-generated pseudo-labels. The architecture of this project shares that skeleton, a frozen proposal generator plus a lightweight module that sorts its output, and differs in the sorting signal: SPOC disambiguates with a learned classifier trained on monocular internet video, whereas EPFL-Smart-Kitchen-30 permits disambiguation from measured geometry, because the rig is calibrated and the hands are tracked in 3D.

TubeletGraph [32] addresses the propagation problem that follows. Given a video and a mask in the first frame, it returns every track descended from that object, including the pieces it breaks into, together with a state graph naming the transformations, assembling frozen components (CropFormer, SAM 2.1, CLIP, an LLM) into a zero-shot system. Its input is exactly the output this project produces, which makes the two complementary rather than competing: the pipeline described here resolves initialisation (which view, which instant, which instance), and TubeletGraph resolves propagation through the transformation.

The limitation these methods share, from the point of view of the framework developed here, is computational. TubeletGraph assembles four large models per frame at a cost on the order of seven seconds per frame, while EPFL-Smart-Kitchen-30 contains 29.7 hours of recording across nine synchronised cameras, roughly 2.9×10^7 frames in total. Processing a single view at that rate would require months of GPU time. Understanding a transformation is intrinsically expensive, so the answer cannot be a cheaper transformation model; it has to be an orchestration that decides where the expensive model is worth running at all. This is what the additional modalities make possible: the action annotations bound when an object is manipulated, and the calibrated rig bounds where it is visible.

2.5 Datasets for hand–object interaction

Several recent datasets combine egocentric or multi-view video with 3D hand annotations, and it is worth being precise about why none of them subsumes the problem addressed here. The decisive point is not any single missing modality but a difference in role: every dataset below supplies object segmentation as ground-truth annotation, or does not supply it at all, and none treats the generation of masks from other modalities as the task itself. EPFL-Smart-Kitchen-30 sits on the opposite side of that line. It is unusually rich in 3D and action priors and contains no object masks whatsoever, which is what makes segmentation conditioned on those priors a task rather than a preprocessing step.

Read row by row, the shortfalls are complementary rather than incidental. HOT3D [2] offers egocentric multi-view with MANO hands, 6DoF object poses and object meshes, but no exocentric views, no body pose and no segmentation task. Ego-Exo4D [15] has the largest

	Ego	Multi-view exo	3D hand	3D body	Object masks	Object pose	Object mesh	Transf. labelled
EgoHOS [37]	✓	×	×	×	✓	×	×	×
VISOR [8]	✓	×	×	×	✓	×	×	×
VOST [33]	(✓)	×	×	×	✓	×	×	✓
HOI4D [22]	✓	×	✓	×	✓	✓	✓	(✓)
HOT3D [2]	✓	×	✓	×	×	✓	✓	×
Ego-Exo4D [15]	✓	✓	✓	✓	✓	×	×	×
SHOW3D [28]	×	✓	✓	×	×	✓	✓	×
EPFL-Smart-Kitchen-30	✓	✓	✓	✓	×	×	×	(✓)

Table 1: Hand-object interaction datasets. (✓) denotes partial coverage: VOST is only partly egocentric, HOI4D labels articulated state rather than material transformation, and EPFL-Smart-Kitchen-30 annotates transforming actions (**cut**, **peel**) without labelling the resulting object states. EPFL-Smart-Kitchen-30 is the only entry combining a calibrated multi-view rig with 3D hand and body pose, and the only one with no object masks, which is the gap this project addresses.

modality overlap with EPFL-Smart-Kitchen-30, providing synchronised ego and exo streams, 3D body and hand pose, object masks, gaze and audio over 1,286 hours. Its masks are annotated in the egocentric view and transferred, however, it carries no object pose or geometry, and it is not organised around fine-grained manipulation of a fixed object set. SHOW3D [28] provides synchronised multi-view with MANO meshes and 6DoF poses at large scale, without segmentation or state. HOI4D [22] is the strongest on the geometry side, with panoptic segmentation, category-level pose, reconstructed meshes and articulated objects, but works from a single egocentric view. VISOR [8] is the closest neighbour in annotation type, providing dense masks and hand-object relations in kitchens, but is monocular and egocentric with no 3D pose of any kind.

Two further requirements of this project appear nowhere in the table: objects that deform and fragment must remain segmentable while they change, and the resulting masks must support a visual 3D reconstruction of the object.

2.6 The EPFL-Smart-Kitchen-30 dataset

Since both projects use it, the properties relevant to them are summarised here.

The recording rig comprises nine static Kinect Azure RGB-D cameras, intrinsically and extrinsically calibrated in a common world frame, together with a HoloLens 2 contributing an egocentric stream with eye gaze and hand tracking. Across 16 subjects and 4 recipes it totals 29.7 hours of recording; the experiments reported here use 49 sessions.

What distinguishes EPFL-Smart-Kitchen-30 from most kitchen datasets is the density and the dimensionality of its annotations. Actions are labelled as segments of the form verb, noun and interval, at approximately 33.8 segments per minute, and include **grab**, **carry**, **put** and **cut** events together with information about which hand performs them and when the hands switch. The kinematic annotations are equally dense: an SMPL body mesh and a MANO hand mesh are fitted at every frame in the same world frame as the cameras, so that annotations, kinematics and calibration all live in a single coordinate system.

What the dataset does not contain is per-pixel object segmentation masks or object identities. Its annotations name objects at the level of the action and never at the level of pixels: a segment records that a bell pepper was cut, not which pixels belong to it, nor which bell pepper it was. Producing those masks, and with them a visual 3D reconstruction of the manipulated objects, is what Project B addresses.

Two consequences of this inventory shape every design decision that follows. First, because the action annotations are per-noun rather than per-instance, they cannot indicate which object is being manipulated when several instances of the same noun are present in the scene. Second, the only per-frame 3D signal available for an object is derived from the tracked hand presumed to hold it, which makes it a proxy whose accuracy varies with the state of the interaction rather than a measured object position. This limitation is revisited in Section 4.3.1

3 STrOMAE : Shared Transformer Orchestrator MAE, a unified framework for multimodal behavior understanding

3.1 Data & Benchmarks (STrOMAE)

3.1.1 MammAlps Dataset (S1 and S2 Releases)

MammAlps [11] is a camera trap corpus recorded in the Swiss National Park, and this project consumes two of its releases, which differ enough that they must be treated as separate benchmarks rather than as more of the same data. The first release defines benchmark 1, behaviour recognition, over clips showing a single animal expressing a single behaviour, each accompanied by a matched audio track and a segmentation map. Three label spaces are supervised simultaneously: species across five classes, activity across eleven, and a multi-label set of nineteen fine-grained actions, of which at most two and usually one are active per clip. The data was randomly split in a train and test set. Segmentation maps ship as class-index arrays at a native resolution of 256 pixels rather than as RGB video, which is why the segmentation embedder is instantiated at that resolution instead of the 224 used for video, and why a one-hot expansion path exists at all. Audio, segmentation maps and video are co-located per track, so all three resolve from the directory of the video path without additional annotation columns. The season-two released under the Prompting Mammalps [12] article. It covers the 2023 and 2024 recording seasons and, for the 2023 season, supplies corrected labels together with additional action, activity and species classes, which makes it a re-annotation of overlapping material rather than purely new material. It is distributed as video only, although each clip retains an embedded audio track. Using that audio as a modality was a testing opportunity rather than a benchmarked experiment.



Figure 1: Samples of the video modality from the Mammalps dataset from S1 and S2

3.1.2 EPFL-Smart-Kitchen-30 Action Recognition Benchmarks

The EPFL-Smart-Kitchen-30 benchmark used here is clip-level joint verb and noun classification over the dense action segments, with a training split of 58,324 clips, which contrasts with MammAlps where one of the three heads is multi-label and where the class counts are an order of magnitude smaller. EPFL-Smart-Kitchen-30 also exercises parts of the framework that MammAlps does not. It supplies fitted body and hand meshes, consumed here as a pose stream, and it offers several synchronised camera views of the same event, consumed as several independent video branches rather than as one stacked input. The configuration used for the multi-view runs declares one branch for the HoloLens view and one for a static side view obtained by string substitution on the video path, with a hand crop at reduced resolution available as a third branch drawn from a separate root directory.



Figure 2: Synchronised EPFL-Smart-Kitchen-30 views consumed as independent branches. Each named branch receives its own embedder and its own positional table, and branches are fused only at the level of the shared trunk.

3.1.3 Annotation format and dataset layout

Both target experiments fine-tuned VideoMAE, but each arrived with its own data loader and its own preprocessing conventions: different annotation schemas, different directory layouts, and different rules for locating the non-video streams. The consequence was that the two pipelines could not share code, and their results could not be compared without first auditing two independently written loaders.

The common contract adopted here is one CSV per split, each row listing a video path, a temporal interval, and one column per supervised head. Intervals may be given in seconds or directly in frame indices, the latter preferred since it bypasses frame-rate arithmetic entirely, which matters for recordings whose nominal and effective rates disagree. Class cardinalities are derived rather than declared: C_h is the number of distinct values found in head h 's column across all present splits, so a mislabelled or absent class surfaces before training rather than silently shrinking an output layer.

What the two datasets do not share is where the non-video streams sit relative to the video. In MammAlps every modality of a track is co-located, so all three resolve from the directory of the video path. In EPFL-Smart-Kitchen-30 the modality directories are siblings of the video directory rather than co-located with it, and the multi-view configuration additionally reaches a separate root for hand crops. The S2 release adds a third case: many clips share a single source video, so no one-to-one mapping from video filename to audio filename exists at all. A single loader serving both experiments must therefore resolve modality paths by more than one rule, which is the mechanism described in Section 3.2.3

Modality	Token count N_m	Value
Video	$(H/P) \cdot (W/P) \cdot (T/t) = 14 \cdot 14 \cdot 8$	1568
Audio	$h \cdot w = 50 \cdot 12$	600
Segmap	$(H/P) \cdot (W/P) \cdot (T/t) = 16 \cdot 16 \cdot 8$	2048
Pose	T	16
V		1568
V+A		2168
V+A+S		4216

Table 2: Fused sequence length $N = \sum_m N_m$ by modality set. H, W : frame height and width (px); T : frames per clip; P : patch side (px); t : tubelet depth; h, w : strided-convolution output dimensions. V: video; A: audio; S: segmentation maps.

3.2 Methods: the STrOMAE framework

3.2.1 Token-level cross-modal fusion

Let $\mathcal{M}$ be the modality set requested for an experiment. Each $m \in \mathcal{M}$ has an embedder E_m mapping its raw input to a token sequence of the shared width d ,

$$E_m : x_m \mapsto z_m \in \mathbb{R}^{N_m \times d},$$

to which a positional table $p_m \in \mathbb{R}^{N_m \times d}$ is added. The embedders are heterogeneous, a 3D convolution for video, a strided 2D convolution for spectrograms, a linear projection for joint coordinates, but their output contract is identical, and that is what makes everything downstream modality-agnostic. The sequences are concatenated,

$$Z^{(0)} = [z_1 + p_1 \parallel \dots \parallel z_{|\mathcal{M}|} + p_{|\mathcal{M}|}] \in \mathbb{R}^{N \times d}, \quad N = \sum_{m \in \mathcal{M}} N_m,$$

and passed through L *shared* blocks, $Z^{(\ell)} = \text{Block}_\ell(Z^{(\ell-1)})$. The pooled representation is the mean over the whole fused sequence followed by a layer norm,

$$\bar{z} = \text{LN}\left(\frac{1}{N} \sum_{n=1}^N Z_n^{(L)}\right),$$

with a class-token variant available as an alternative.

The decision that matters is that self-attention in every block runs over the whole concatenated sequence. A video token can attend to a specific mel band or a specific hand joint at any depth, rather than meeting it only after both have been pooled to a vector. The cost is that adding a modality grows N , and attention is $\mathcal{O}(N^2)$, so cost scales quadratically in the fused length rather than linearly in the number of branches. Table 2 makes this concrete: moving from V to V+A lengthens the sequence by 38% and multiplies attention cost by $(2168/1568)^2 \approx 1.91$. This is why gradient checkpointing is enabled for the ViT-Large configurations and why the per-device batch in Section 3.3 is only four.

3.2.2 Multi-task heads and loss orchestration

An arbitrary head set $\mathcal{H}$ reads the same $\bar{z}$, each head with its own MLP, metric type and loss. The objective is

$$\mathcal{L} = \sum_{h \in \mathcal{H}} \lambda_h \mathcal{L}_h,$$

with λ_h exposed in the configuration. For a multiclass head with logits $\hat{y} \in \mathbb{R}^{C_h}$, one-hot target y and smoothing ε ,

$$\mathcal{L}_h = - \sum_c \tilde{y}_c \log \text{softmax}(\hat{y})_c, \quad \tilde{y}_c = (1 - \varepsilon) y_c + \varepsilon / C_h.$$

The multi-label action head of MammAlps is the interesting case, and the framework supports two formulations. Binary cross-entropy treats the C_h classes as independent decisions,

$$\mathcal{L}_h^{\text{BCE}} = -\frac{1}{C_h} \sum_c \left[y_c \log \sigma(\hat{y}_c) + (1 - y_c) \log (1 - \sigma(\hat{y}_c)) \right],$$

whereas soft-target cross-entropy first normalises the multi-hot target into a distribution and then applies the softmax loss,

$$\tilde{y} = y / \|y\|_1, \quad \mathcal{L}_h^{\text{sCE}} = - \sum_c \tilde{y}_c \log \text{softmax}(\hat{y})_c.$$

The distinction is visible in the gradients. For BCE, $\partial \mathcal{L} / \partial \hat{y}_c = (\sigma(\hat{y}_c) - y_c) / C_h$, so all C_h logits receive gradient independently and, with one active label out of nineteen, eighteen of them are being pushed toward zero at every step. For the softmax form, $\partial \mathcal{L} / \partial \hat{y}_c = p_c - \tilde{y}_c$, which sums to zero across classes: the update *redistributes* mass toward the correct label rather than issuing C_h independent pushes. Since MammAlps actions are at most two and usually one per clip, the labels genuinely compete for a single slot, and the softmax form is the better match. Binary cross-entropy remains the framework default, since it is correct whenever labels co-occur freely; the softmax form is selected per head and never applied implicitly.

Class imbalance is handled by the sampler rather than by loss weighting. With N training clips and class frequency f_c , the per-class weight and per-sample weight are

$$w_c = \frac{N}{f_c + \lambda N}, \quad w_i = \sum_{h \in \mathcal{H}} \sum_{c: y_{i,c}^h = 1} w_c,$$

normalised by $\max_i w_i$. The smoothing term λ caps the oversampling of the rarest classes at roughly $1/\lambda$; the reported runs use $\lambda = 10^{-3}$, matching the reference. Two properties are deliberate. First, the sum runs over *all* heads, so a clip common in species but rare in action is still upsampled; an earlier version summed over the first head only and left the remaining heads uncorrected. Second, inverse-frequency loss weighting is disabled whenever the balanced sampler is active, since applying both compounds two corrections for the same imbalance.

3.2.3 Declarative configuration

Everything described above, the active modality set, their per-branch parameters, the head set and the weights λ_h , is selected by configuration rather than by code. Two objects with disjoint responsibilities drive it. A `DatasetConfig` answers what the data is; a `TrainingConfig` answers how to train on it, and carries every hyperparameter of the optimiser, schedule and augmentation. A complete experiment is the pair, and switching from EPFL-Smart-Kitchen-30 to MammAlps changes only the first.

```
DatasetConfig(
    name="my_dataset",
    modalities=["video", "audio"],
    modality_params={
        "video": {"input_size": 224},
        "audio": {"input_size": 224, "n_mels": 128, "audio_length": 512},
    },
    heads={
        "verb": {"metric": "multiclass"}, "objects": {"metric": "multilabel"},
    },
    head_weights={"verb": 1.0, "objects": 2.0},
    video_path="...", anno_path="...",
)
```

The path divergence noted in Section 3.1.3 is absorbed by three resolution rules declared on the same object. An explicit CSV column per modality removes all ambiguity and always takes precedence. A `dirname` rule reuses `dir(video_path)`, which serves MammAlps. A `subject_session` rule takes the first two path components as a shared root, which serves EPFL-Smart-Kitchen-30. For the S2 case where clips share a source video, an `audio_path` column naming an exact file overrides the derived name entirely. These four cases cover both experiments, and adding a dataset whose layout matches none of them requires a new rule rather than a new loader.

The resulting user surface reduces to three names. One call loads a configuration file and returns a handle, one method overrides hyperparameters, and one launches a run. A hyperparameter sweep is therefore a loop over the same handle, with each launch writing to its own numbered output directory beneath the configured root so that successive runs cannot overwrite one another’s checkpoints.

```
import stromae

opt = stromae.init(path_config="run.yaml", dataset_name="ds1")
for lr in [1e-4, 5e-5, 1e-5]:
    opt.update_params(lr=lr)
    results.append(opt.finetune())
```

3.2.4 The assembled framework

Figure 3 shows the components of the preceding four sections wired together. Two paths run through the system and should be read separately. Configuration flows from a single file into the choice of which embedders exist, which heads are attached, how modality paths resolve, and how the optimiser is scheduled. Data flows from the annotation CSVs through those embedders into one concatenated sequence, through the shared trunk, and out to the heads. Nothing on the data path is dataset-specific: replacing EPFL-Smart-Kitchen-30 with MammAlps changes only which embedders the configuration instantiates and which heads it declares, while the trunk, the concatenation and the loss aggregation are untouched.

3.2.5 Validation protocol and defects corrected

A framework of this kind fails in a characteristic way: it runs, the loss decreases a little, the metrics are mediocre, and nobody can say whether the model is weak or the plumbing is broken. I therefore imposed a hard entry criterion before any full run: *the model must drive the loss to approximately zero on a batch of four samples*. If it cannot memorise four examples, no number obtained on the full dataset is interpretable.

The first attempts failed, and informatively. On a multilabel configuration the loss stalled at ≈ 0.5623 while gradient norms stayed healthy at ≈ 3.5 , ruling out a detached graph or dead activations. That value is not arbitrary. A constant predictor emitting the marginal label rate p for every class attains exactly the binary entropy

$$\mathcal{L} = H(p) = -(p \log p + (1 - p) \log(1 - p)),$$

and at the marginal rate $p \approx 0.25$ of that configuration, $H(0.25) = 0.3466 + 0.2158 = 0.5623$. The model was predicting the prior and nothing else. A loss pinned to a closed-form constant is a much stronger diagnostic than a merely high loss, because it identifies *which* degenerate solution has been reached. The audit that followed produced Table 3.

3.2.6 Distributed training

Training runs under PyTorch DDP, one process per device, on H100 nodes with datasets mounted from a persistent volume. Because token count rather than parameter count grows with each

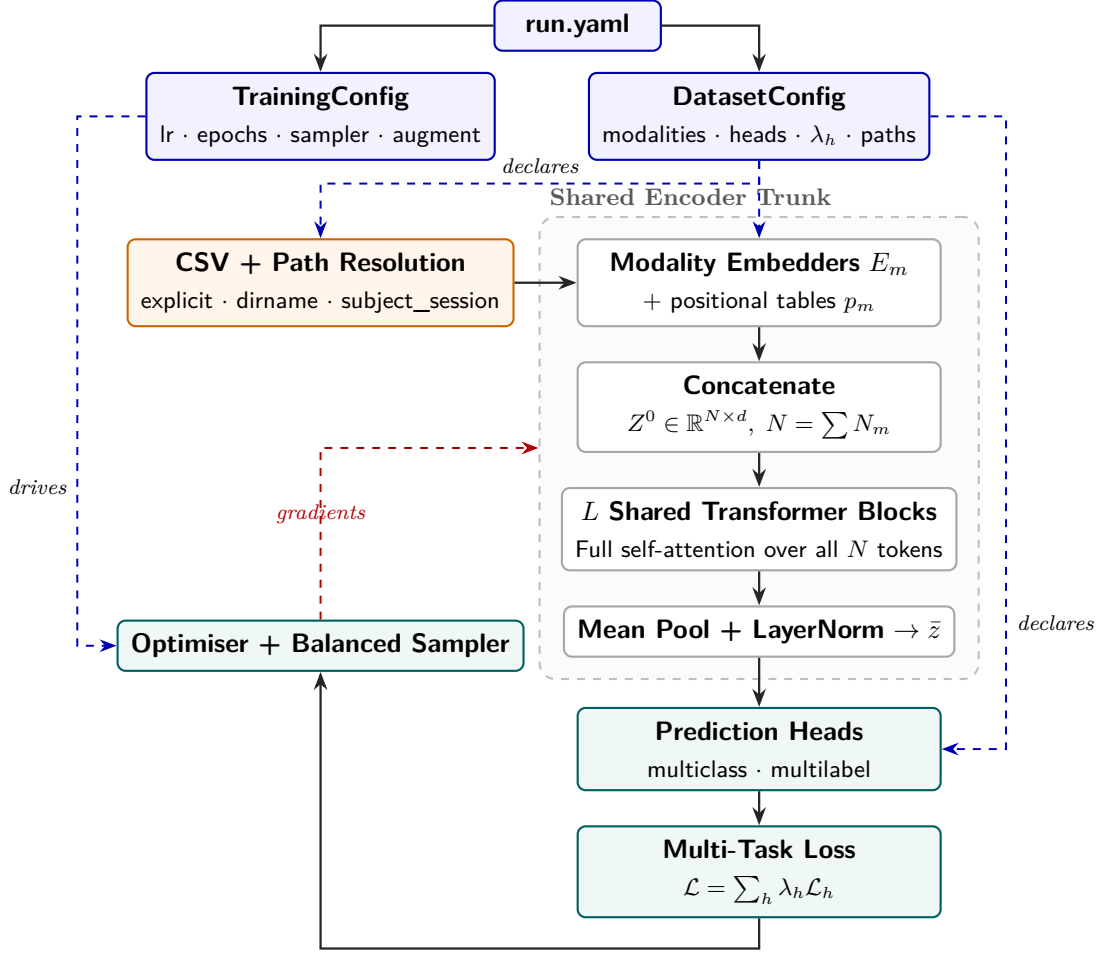


Figure 3: System architecture and declarative execution pipeline of the unified STrOMAE framework.

Defect	Consequence and correction
Loss selected by metric, not by declared loss	The criterion was built from <code>head_losses</code> but the training branch was chosen by the head’s <code>metric</code> , so cross-entropy on a multilabel head ran through the BCE path and the loss rose. Dispatch now keys on the criterion actually constructed.
Learning rate rescaled by batch	The passed rate is multiplied by $B_{\text{eff}}/256$ before use. At $B_{\text{eff}} = 16$ this divides it by 16, so a published rate transcribed directly under-trains by an order of magnitude, presenting as a plausible but misleading run rather than a crash.
Double imbalance correction	Balanced sampling and inverse-frequency loss weights compound. Weighting is disabled for the reported runs.
Sampler balanced one head	Per-sample weights summed over the first head only, leaving clips rare in the remaining heads under-sampled.
Class weights on host memory	Weights were never moved to the compute device, raising a device mismatch as soon as weighting was active on a GPU.
Duplicated declarations	Input resolution and frame count were declared both as flags and in the dataset config; a third quantity proved to be dead code carrying two mutually inconsistent defaults.
Audio filename derivation	Deriving the audio name from the video stem collides when clips share a source video. An explicit per-row path now takes precedence.
Argmax-only evaluation	Reporting assumed single-label prediction, which is uninformative for a multilabel head. Per-head and overall mAP were added, gathered across ranks.

Table 3: Defects isolated by the four-sample criterion and the audit it prompted. The first two change how any result recorded before their correction should be read.

modality, the per-device batch is small and the effective batch is recovered by accumulation:

$$B_{\text{eff}} = B_{\text{dev}} \cdot A \cdot W = 4 \cdot 2 \cdot 2 = 16.$$

The optimiser then receives

$$\text{lr}_{\text{used}} = \text{lr}_{\text{passed}} \cdot \frac{B_{\text{eff}}}{256} = \frac{\text{lr}_{\text{passed}}}{16},$$

so obtaining an effective 10^{-5} requires passing 1.6×10^{-4} . This rescaling is invisible in a launch script and was the single most consequential of the defects in Table 3. Evaluation can also be distributed, at the cost of duplicated samples when the validation set does not divide evenly across ranks. Gradient checkpointing is enabled for ViT-Large.

3.3 Experiments & Results (STrOMAE)

Two benchmarks were reproduced in parallel through the same codebase, differing only in configuration: MammAlps benchmark 1 and the EPFL-Smart-Kitchen-30 verb and noun task. They are reported on equal footing, since the purpose of running both is to establish that a single framework reaches each dataset’s published baseline without dataset-specific code, not to treat one as primary and the other as a transfer check. Their protocols differ in every respect that matters, which is what makes the pair informative: MammAlps supervises three heads including a multi-label one and is scored by macro mean average precision, whereas EPFL-Smart-Kitchen-30 supervises two single-label heads and is scored by top-1 and top-5 accuracy.

Both use a ViT-Large backbone and the balanced sampler of Section 3.2.2. MammAlps runs are trained for 150 epochs from a Kinetics-700 initialisation at an effective batch of sixteen,

with sixteen frames sampled inside a random five second window, and are scored per class and averaged over the classes present in the test split. EPFL-Smart-Kitchen-30 runs are evaluated on the full test split of 35,032 clips at the best checkpoint rather than the final epoch. In both tables the published values are reproduced alongside, so each cell reads as a comparison rather than as an isolated score.

3.3.1 Sanity Convergence & Impact of Corrected Defects

The controlled over-fitting protocol was the mechanism through which most of Table 3 surfaced, and two of its findings dominate everything that follows. The first is the learning rate rescaling. At an effective batch of sixteen the internal factor is one sixteenth, so a rate transcribed directly from a published recipe trains at roughly an order of magnitude below the intended value, which presents not as a crash but as a run that converges to a plausible and entirely misleading score. The second is the loss dispatch defect, whose signature is visible in the training curve rather than only in the final number: a cross-entropy criterion applied through the multi-label branch produced a loss that rose over training on the action head while the single-label heads behaved normally, which is why the protocol caught it and a full run would likely not have.

Capacity was isolated separately. Table 4 reports an earlier configuration using a ViT-Base backbone at 100 epochs, which falls short of the published benchmark on every head and both modality sets, by roughly one tenth of a point on average. The same pipeline at ViT-Large reaches or exceeds the benchmark throughout, as Section 3.3.2 shows, which places backbone capacity rather than any pipeline detail as the single largest factor separating the two regimes.

Head	V+A		V+A+S	
	Ours	Paper	Ours	Paper
Actions	0.407	0.463	0.323	0.417
Activity	0.428	0.475	0.331	0.452
Species	0.270	0.473	0.432	0.531
Average	0.368	0.470	0.362	0.467

Table 4: Macro mAP on MammAlps benchmark 1 with a ViT-Base backbone at 100 epochs, the exploratory configuration preceding the reported runs. Every cell falls below the published benchmark.

3.3.2 MammAlps Multimodal Species and Activity Recognition

Table 5 reports the ViT-Large runs against the published benchmark across the three modality sets. Every one of the twelve comparisons is favourable, and the margin varies systematically by head: the activity head gains most, by 0.168 on video alone, the action head gains moderately, and the species head gains least, by 0.023 on video alone. The best configuration overall is video with audio, which exceeds the published average by 0.098.

Exceeding a published benchmark on a reproduction isn’t great news, it raises questions about the validity of the data pipeline and the protocol deviations listed in Section 2.1 apply in full here. The backbone is Kinetics-700 fine-tuned, which is a more favourable initialisation than a plain masked-pretrained checkpoint. The margin should therefore be read as evidence that the unified framework does not degrade dataset-specific behaviour, which was the stated goal, rather than as a claim of methodological improvement.

The per-class breakdown shows that the residual error is not spread evenly. On the best configuration the action head ranges from 0.933 on shaking fur and 0.910 on walking down to 0.001 on urinating and 0.004 on defecating, and the species head is near-perfect on wolf

Head	V		V+A		V+A+S	
	Ours	Paper	Ours	Paper	Ours	Paper
Species	0.518	0.495	0.570	0.473	0.545	0.531
Activity	0.596	0.428	0.611	0.475	0.520	0.452
Actions	0.515	0.458	0.522	0.463	0.430	0.417
Average	0.543	0.460	0.568	0.470	0.499	0.467

Table 5: Macro mAP on MammAlps benchmark 1, ViT-Large, 150 epochs. Published values are given alongside for direct comparison.

and red deer while reaching 0.001 on hare. The failures concentrate almost entirely on classes with support of ten examples or fewer in the test split, which are close to unlearnable from that many instances and which the original benchmark also misses. The framework’s headline averages are therefore driven by the frequent classes, and any future gain on this benchmark will have to come from the tail.

A second comparison, in Table 6, applies the same video-only configuration to the corrected season-two annotations, which fix labelling errors and introduce additional action, activity and species classes. Species improves slightly while activity and actions both fall by more than 0.13. The additional classes are themselves rare, so part of the drop is the tail effect described above operating on a longer tail, but the magnitude suggests the corrected labels also removed some signal the earlier annotations had made artificially easy.

Head	2023 v2 (V)	Δ vs. v1 (V)
Species	0.535	+0.017
Activity	0.463	−0.133
Actions	0.358	−0.157
Average	0.452	−0.091

Table 6: Macro mAP on the corrected season-two 2023 annotations, video only, ViT-Large, 150 epochs, compared against the video-only column of Table 5.

3.3.3 Modality Ablation Studies

The modality axis of Table 5 is an ablation in its own right, since the schedule is held fixed and only the declared modality set changes, which the configuration interface makes a one-line edit. Adding audio to video improves every head and lifts the average by 0.025, which is consistent with the ecological content of the audio track and with the same ordering reported by the reference. Adding segmentation maps on top of that reverses the gain and costs 0.069 on the average, placing the three-modality configuration below video alone.

That reversal is larger here than in the published results, where the two configurations are nearly indistinguishable, and the asymmetry described in Section 2.1 is the most plausible explanation. The segmentation embedder is randomly initialised, it operates on class-index arrays at a different native resolution from the video stream, and it lengthens the fused sequence by a third without contributing pretrained structure, so it imposes an optimisation cost that its signal does not repay at this data scale. The same reasoning predicts that a segmentation branch initialised from a pretrained checkpoint, or a longer schedule, would narrow the gap, and neither was tested here. Even though the theory of having a pretrained backbone agnostic to modalities was appealing, it seemed like too much of a computational reach.

3.3.4 EPFL-Smart-Kitchen-30 verb and noun recognition

The second benchmark supervises verb and noun jointly, with 31 and 78 classes, over four streams declared in a single configuration: the egocentric HoloLens view, a static exocentric view obtained by string substitution on the video path, and 3D body and hand pose. The backbone is ViT-Large, 309M parameters. Nothing outside the configuration file differs from the MammAlps runs of Section 3.3.2, which is the point: the two benchmarks differ in modality set, head structure, label cardinality and scoring metric, and are served by the same model and the same training loop.

Head	Ours		Paper	
	Top-1	Top-5	Top-1	Top-5
Verb	62.25	93.22	61.04	93.59
Noun	53.72	79.79	50.94	77.52

Table 7: Top-1 and top-5 accuracy on the EPFL-Smart-Kitchen-30 benchmark, ViT-Large, egocentric and exocentric views with 3D body and hand pose, full test split. The higher value of each pair is shown in bold.

The verb top-1 score of 62.25 exceeds every row of the published table, including the row that adds eye gaze and hand-cropped views on top of the four streams used here. Noun top-1 at 53.72 is above all rows except the full modality combination. Both top-5 scores sit mid-range. Read together, this places the unified framework at parity with, or slightly above, the published baseline on the modality set it can express, which is the claim this experiment exists to support. Three deviations from the published protocol must be stated. First, the benchmark supervises three heads, action in addition to verb and noun, whereas the configuration used here declares only two. Under the EPIC-KITCHENS convention that EPFL-Smart-Kitchen-30 follows, action is the composed (verb, noun) pair rather than an independent annotation, so it is derivable from the per-sample predictions already written by the evaluation. Second, the published table reports separate scores over head and tail class populations, a long-tail split whose class partition is not reproduced here, so only the all-classes column is comparable. Third, the strongest published rows use eye gaze, a modality the framework does not implement, so no comparison against those rows is claimed.

3.3.5 Across the two benchmarks

The two reproductions land differently, and the difference is informative. On MammAlps the ViT-Large runs exceed the published values on all twelve comparisons, whereas on EPFL-Smart-Kitchen-30 they sit at parity, above the reference row on both top-1 scores and marginally below on verb top-5. A framework that systematically beat both would be more suspicious than one that reaches each baseline by a different margin, since the two datasets share no data, no label structure and no metric.

What transferred between them was not performance but failure modes. Every defect in Table 3 was dataset-agnostic: the learning rate rescaling, the loss dispatch keyed on the wrong attribute and the double imbalance correction each applied identically to both pipelines once found, and each was found on one dataset and fixed for both. That is the concrete return on unification, and it is not visible in either results table.

Backbone capacity dominated everything else. The same pipeline at ViT-Base fell below the MammAlps benchmark on every head and both modality sets, while at ViT-Large it exceeded it throughout. No configuration change produced a comparable shift.

4 Interaction-Aware Object Segmentation on EPFL-Smart-Kitchen-30

4.1 Data Characterisation and Sensor Bottlenecks

4.1.1 Kinematic Streams and Their Failure Envelope

We established that the only per-frame three-dimensional signal available for an object is derived from the tracked hand presumed to hold it. That limitation is developed here, because its structure rather than its existence is what constrains the pipeline.

The file `object_pose.json` does not contain a measured object position. It contains the position of the hand presumed to be holding the object, obtained by clustering MANO wrist and joint positions, and produced by a model developed earlier within the group rather than by any observation of the object itself. The accuracy of this proxy is not a single number but a function of the interaction state, and Table 8 gives the envelope.

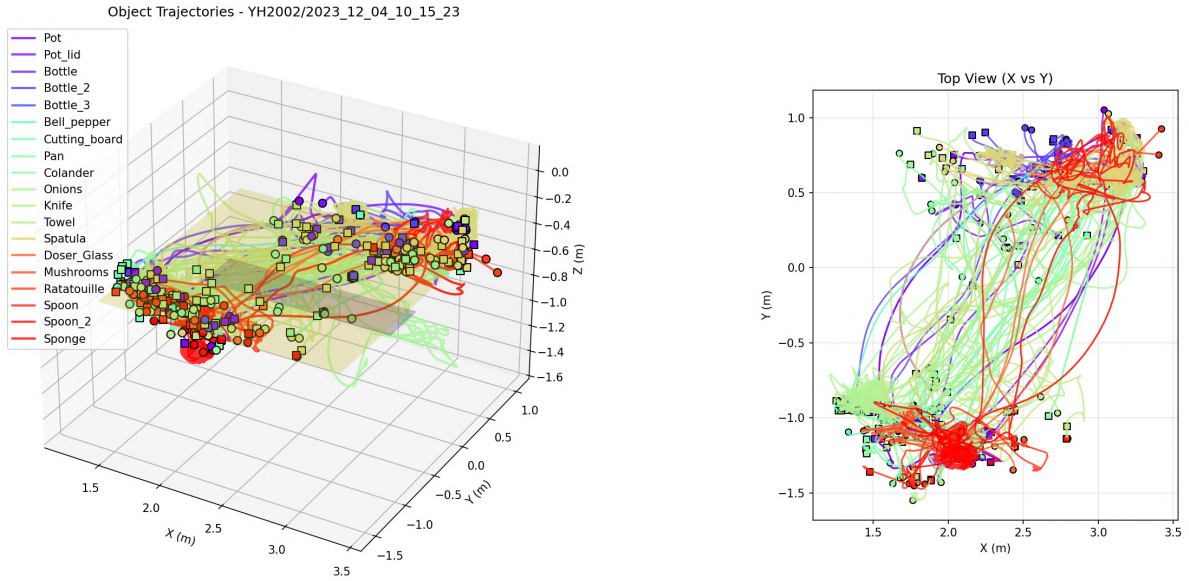


Figure 4: Objects pose derived from hand positions on a single session

Interaction state	Behaviour of the hand-derived proxy
Held	Offset from the true object centre by the grip distance, on the order of 5 to 15 cm. Usable for cropping.
At rest	We only record where the object was left without updating the position until it is picked up again from dense annotations.
Fragmented	Meaningless, there is no need for the hand position when the object has already gone through transformations (cut in pieces, peeled etc.)

Table 8: State dependence of the object position proxy. The degradation is ordered against the difficulty of the segmentation problem: the proxy is least reliable exactly where segmentation is hardest.

The consequence that recurs throughout this part is that no geometric test in the pipeline is ever really a test of proximity to the object. It is a test of proximity to the hand, because there is no signal about the object that is independent of the hand. A geometric veto is therefore

weakest during cutting, when the fragments lie on the board while the hand is occupied with the knife, which is precisely the regime in which such a veto would be most valuable.

A last property of the annotations bounds what any method can be asked to resolve. Action annotations are recorded per noun rather than per instance, so when several instances of a noun are present the annotation cannot indicate which one is being manipulated. Nothing in the released data disambiguates them, which means instance-level identity, if it is wanted at all, has to come from geometry rather than from the labels.

4.1.2 The Absence of Usable Depth

The rig is described as RGB-D, which invites the assumption that metric depth is available in every view. It is not. The export used throughout this project contains no usable depth stream. The consequence is geometric and it propagates through everything downstream. A mask in a calibrated view, combined with the intrinsics and extrinsics, does not determine a point in space. It determines a ray. Moreover access to depth data would have significantly facilitated 3D reconstruction, or at least more precise 3D object positions across the multi-view setup.

This forecloses the strategy that would otherwise be the natural response to occlusion, and which was proposed as a first idea early in the project: segment the object in a camera that sees it clearly, recover its 3D coordinates, and project the result into the view where it is occluded. That requires depth and cannot be implemented here. What remains is the epipolar constraint between pairs of calibrated views and triangulation from two or more, both of which require the object to be segmented simultaneously in several cameras. Multi-view processing is therefore not a way of improving a result obtainable from one camera. It is the only way of obtaining the result at all.

4.1.3 Evaluation and Failure Modes of Sparse-View Reconstructors

The project began from the opposite end. Its initial framing was to recover 3D meshes of the kitchen objects so that they could be projected onto the video and tracked, which admitted two routes: reconstruct geometry from the available views, or fit pre-existing templates to 2D masks. The first was attempted, both because it would yield object models specific to this environment and because the dataset is large enough that abundance of data seemed likely to compensate for the absence of an object-centric capture protocol.

The strategy adopted was a compromise between the two. Rather than reconstructing objects during manipulation, when hands occlude them and motion blurs them, each object would be reconstructed once at rest, before its first grab event, yielding a static template to be fitted to per-frame masks thereafter. The object was localised using the hand centroid immediately preceding the first grab, the images were cropped around that centroid so that the reconstruction would concentrate on the object rather than the room, and the crops were passed to VGGT [35], with SAM 3 used to isolate the object within the resulting point cloud.

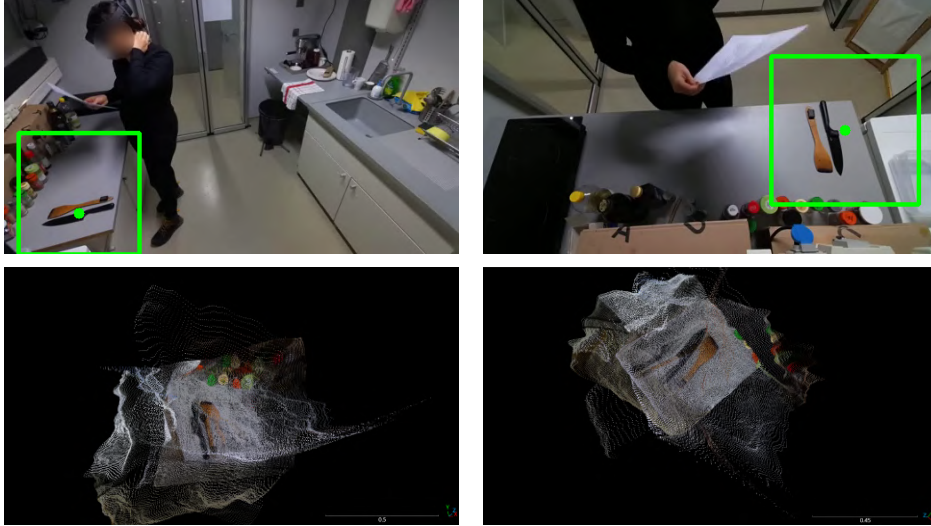


Figure 5: Sparse-view reconstruction of a knife and spatula from hand-centroid crops. Neither reconstructor recovers usable object geometry.

The attempt failed, and the reasons generalise beyond the particular models. VGGT infers camera positions as part of its forward pass and back-projects using those inferred poses, without consuming the Kinect intrinsics the rig already provides. On a calibrated rig this is not merely redundant but harmful, since the reconstruction is expressed in a geometry the model has invented rather than the one that was measured. Compounding this, the objects are physically distant and occupy a small fraction of each frame, which is what made cropping necessary; but cropping a 512×288 frame around a distant object leaves very little resolution to work with. A super-resolution step before the model was considered and rejected, on the grounds that it would not address the back-projection problem, which is the more fundamental of the two.

MASt3R [19] was then evaluated on the same task, on the expectation that its explicit pairwise matching between views would extract more from a small view set than VGGT’s joint formulation. It performed notably worse, which is consistent with its design: MASt3R is intended to exploit a large set of view pairs, and with a small number of widely separated cameras observing a distant object there are few reliable correspondences for the matching stage to find. Both belong to the same family of scene-level reconstructors that jointly estimate pose and geometry, and both are therefore mismatched to a regime in which pose is already known exactly and the target is small, distant and often specular.

The conclusion drawn in discussion with my supervisor was to abandon object-centric reconstruction and use existing templates for these common kitchen objects, on the reasoning that the scientific goal is to characterise how the subject responds to object positions and states rather than to produce faithful object geometry.

That decision determines the rest of this part. If templates supply the geometry, the quantity that must be recovered from video is the per-frame mask, and segmentation moves from being a preprocessing step to being the deliverable.

4.2 Methods: Benchmarking and Geometry-Guided Orchestration

4.2.1 The Interaction Module and Why It Was Benchmarked First

Segmentation of manipulated objects was approached from the assumption, natural and widely shared, that manipulation and occlusion would be the dominant difficulties. Under that assumption the appropriate response is to inject information about the interaction into the segmenter, and the design pursued was a module in the spirit of CaRe-Ego [31], whose Hand-

guided Object Feature Enhancer conditions object features on hand features through cross-attention rather than concatenating the two.

Creating enriched embeddings in an interaction aware way was the intuition to help the decoder make fine-grained segmentations. Moreover, the dataset at hand allows us to leverage other modalities like 3D MANO data. Analysis of the SAM 3 architecture [4] identified two viable injection points, under the self-imposed constraint that the decoder must not be replaced and that the contribution should refine the inputs rather than reimplement the model. Figure 6 locates them.

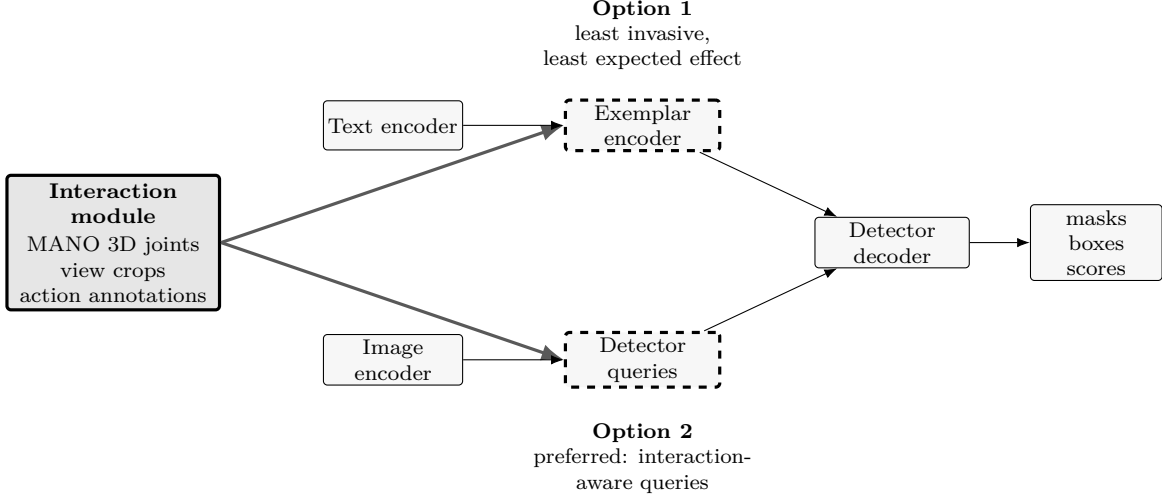


Figure 6: The two candidate injection points for the interaction module into a frozen SAM 3. Dashed boxes are the entry points considered; the decoder is not modified in either case. The second was preferred as the better balance between the extent of the modification and its likely effect on the output.

The module was implemented. Training it, however, requires ground-truth masks of manipulated objects, which EPFL-Smart-Kitchen-30 does not provide, and this is where the plan met its real cost. Table 9 sets out the three levels of annotation effort that were envisaged.

Strategy	Cost and risk
1 Segment one exocentric view manually, reproject onto the rest, hold out the egocentric view to measure transfer	Cheapest, but presupposes the mask re-projection that Section 4.1.2 shows to be impossible without depth
2 As above, followed by fine-tuning on a small manually segmented set	Moderate; inherits the same reprojection dependency
3 Full training on a manually segmented dataset	Weeks of manual annotation, and the only option free of the above

Table 9: Annotation strategies considered for training the interaction module.

To keep the annotation budget productive, frames were to be selected by farthest point sampling in three dimensions, beginning from the first active frame of an event and repeatedly adding the frame whose 3D object position lies farthest from all frames already chosen. Given a set S of already-selected frames and candidate positions p_t ,

$$t^* = \arg \max_{t \notin S} \min_{s \in S} \|p_t - p_s\|_2,$$

with a per-event budget proportional to trajectory length and capped so that one long event cannot dominate the set. The rationale is that two frames close together in 3D look similar

from every camera whereas two frames far apart are genuinely different, so this selection covers pickup, carry and putdown without requiring any of those stages to be identified in advance. On a 200-frame carry event, five frames chosen this way recover the object on the table, just lifted, mid-carry, near the destination and set down. Annotation was begun in CVAT with auxiliary labels recording which hand is interacting, the level of occlusion on a scale of none, partial and heavy, and the line of contact between hand and object.

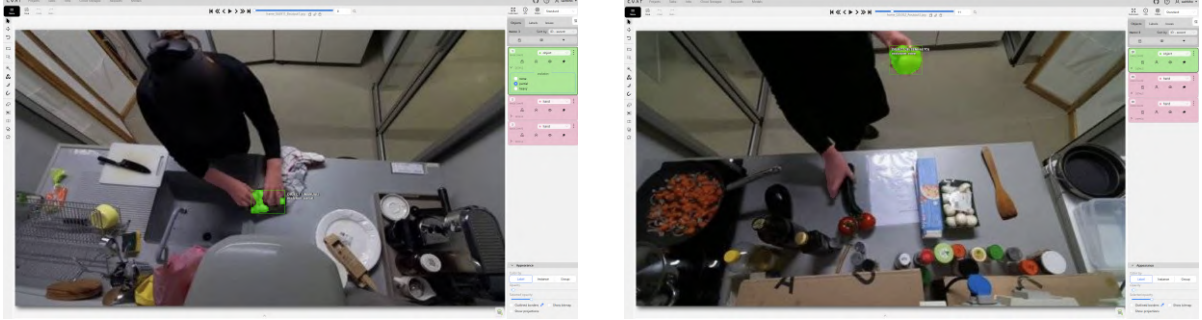


Figure 7: CVAT platform for manual annotation

At this point the cost of proceeding became explicit. Producing a training set of the required size is a matter of weeks, and the entire expenditure is justified only if the module addresses SAM 3’s actual weaknesses. That conditional had never been tested. The decision to benchmark SAM 3 before continuing with the annotation is the decision this part of the report is organised around, and it is defended not by its outcome but by its asymmetry: the benchmark cost days and the annotation would have cost weeks.

4.2.2 Benchmark Protocol

The benchmark separates two factors that are ordinarily conflated. Twenty-four clips of thirty seconds were drawn from the working session YH2002/2023_12_04_10_15_23 and assigned to three conditions, as laid out in Table 10.

	Condition	Manipulated	Transformed	Isolates
1	Present, not handled	no	no	Baseline recognition and tracking
2	Handled, identity preserved	yes	no	The effect of manipulation alone
3	Transformed by the action	yes	yes	The effect of state change

Table 10: Benchmark conditions. Separating rows 2 and 3 is the point of the design: if interaction is the difficulty, performance falls between rows 1 and 2; if state change is the difficulty, it falls only at row 3. A single undifferentiated category of interaction would make the two indistinguishable.

The verb vocabulary used to assign clips to Condition 3 is a closed set consisting of Cut, Peel, Grate, Split and Sauté (for this given session).

Each clip was processed with a text prompt naming the object, and every masklet returned was persisted as a run-length encoding in JSONL so that scoring and analysis could be repeated without re-running inference. Scoring is a per-clip visual assessment of the proportion of frames on which the correct object is correctly segmented, recorded alongside a description of the failure. This is a qualitative measure and not an intersection over union against ground truth,

for the reason that motivated the benchmark in the first place: no ground truth exists. This is an empirical way of guiding the project towards a task-oriented solution.

4.2.3 The Orchestration Pipeline

The pipeline described here is the one the benchmark results of Section 4.3 motivate, and it is presented before those results only because the two must be read together. Its governing principle is that SAM 3 proposes masklets and geometry disposes: the segmenter is treated as a generator of candidates whose recognition decisions are not trusted, and the calibrated rig is treated as the arbiter of which candidates are admissible. Figure 8 shows the assembled system.

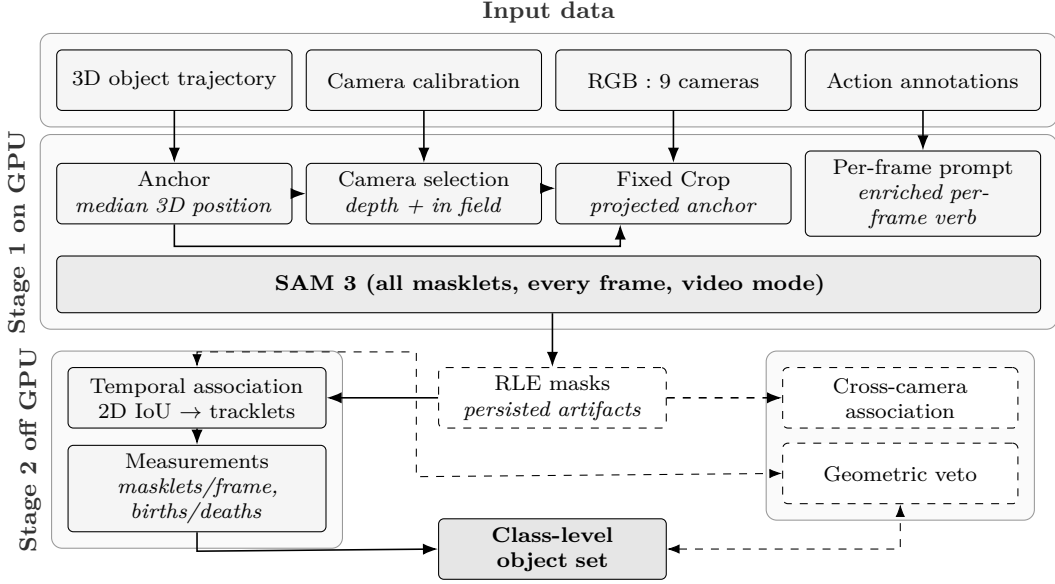


Figure 8: The orchestration pipeline. Solid boxes are implemented; dashed boxes are measured but not applied (geometric veto) or not implemented (cross-camera association).

Activity gating. The action annotations merge padded action intervals into events per object. Outside its events an object is at rest, and the mask computed at the start of that span is carried forward as a world-state mask, retained even when a person walks between the object and the camera, on the reasoning that the camera is not supposed to see the object during that interval and its absence from the image is therefore not evidence about the object. Beyond correctness, this gating is what makes the pipeline affordable, and Section 4.4 quantifies that.

Window structure. Because the SAM 3 video memory bank grows without bound, sessions are reset every 300 frames. Events are cut into windows aligned with this period, so the reset boundary is a designed feature of the schedule rather than an interruption of it. This is one of the biggest bottlenecks because SAM 3’s memory bank being reset means the masklets are all lost and need to be re-detected based on the initial prompt. This is why the enriched per-frame verb prompt was proposed, unfortunately, a new prompt cannot be injected without creating a new instance. Therefore, this pipeline was quickly in jeopardy.

4.2.4 Camera Selection

The selection rule that produced the results reported here is geometric and deliberately simple. At each frame the tracked anchor position X_t of an object is projected into every calibrated camera. Writing $[R_c | t_c]$ for the world to camera transform and K_c for the intrinsics of camera

c , scaled from the calibration resolution to the resolution of the exported clips,

$$\tilde{u} = K_c (R_c X_t + t_c), \quad (u, v) = \left(\frac{\tilde{u}_1}{\tilde{u}_3}, \frac{\tilde{u}_2}{\tilde{u}_3} \right), \quad z_c(t) = \tilde{u}_3.$$

A camera is a candidate at frame t when the projection falls inside the image, $0 \leq u < W_c$ and $0 \leq v < H_c$, and when the object is close enough, $0 < z_c(t) \leq z_{\max}$, with $z_{\max}$ set to two meters. Candidates are then ranked by depth and the nearest is selected.

Because a frame-by-frame argmin over depth produces frequent switching, and because every switch discards the SAM 3 memory bank, the choice is smoothed by hysteresis: the current camera is retained unless it stops being a candidate, or unless a competitor is closer by more than a fixed fraction for a minimum number of consecutive frames. Segments shorter than a minimum length are absorbed into their neighbour.

Two limitations of this rule are worth stating, because they motivate the redesign described next rather than being incidental defects.

The first is that depth is a proxy for visibility, and the frustum test only establishes that the object lies within the field of view. Neither says anything about occlusion by the operator’s body, by the counter, or by a cabinet door. This matters more than it might appear, because the anchor being projected is derived from the tracked hand, so the principal occluder during manipulation is frequently the very arm holding the object. The candidate set is therefore systematically optimistic in exactly the circumstances the pipeline exists to handle.

The second is that the hysteresis optimises for something other than the stated goal. Its purpose is to minimise camera switches, and the reason switches are costly is that they destroy temporal memory, which is a tracking concern. One switch in the working session fired 44 frames after the object had become poorly visible in the outgoing view, a delay attributable to the hysteresis threshold rather than to the geometry. More fundamentally, a rule that selects a single best view cannot serve a reconstruction objective at all, since a mask in one calibrated view constrains the object to a ray and not to a point.

4.2.5 A Multi-View Redesign

Recovering a three-dimensional reconstruction requires the object to be segmented in at least three views at the same instant. Two views suffice to intersect rays, but if there’s a disagreement we wouldn’t be able to resolve it.

The redesign replaces the single selected camera by a probe. Each event is cut into windows aligned with the memory-bank reset that must happen anyway, so that re-selecting cameras costs nothing in tracking continuity.

The scoring rule is the part that required rethinking. Apparent area in pixels is not a measure of visibility: for an object of true cross-section A at depth z in a camera of focal length f_x , the projected area is

$$A_{\text{px}} = A \left(\frac{f_x}{z} \right)^2,$$

so the measured area depends as much on the camera’s distance and focal length as on how much of the object can actually be seen. Evaluating this expression across the nine cameras of the rig for a fixed A shows how large the effect is: the same object covers roughly a hundred times more pixels in the nearest camera than in the furthest, even when neither is blocked. Thresholding raw area would therefore mostly repeat the depth ranking instead of adding to it. Dividing out the distance and the focal length gives a score,

$$\hat{A}_c = A_{\text{px},c} \left(\frac{z_c}{f_{x,c}} \right)^2,$$

that does not depend on how far the camera is, what lens it has, or what resolution it records at. A view whose score is much lower than the best one is therefore blocked rather than simply far

away. Occlusion, which the geometric tests cannot detect, is in this way read off what SAM 3 returns.

The probe additionally provides a synchronised multi-view observation at no extra cost, since the same frame is segmented in every candidate view. Rays through the mask centroids should intersect, which allows a view that has segmented a different object to be identified and dropped before it consumes any tracking time.

This redesign is implemented and its geometry has been verified against the calibration in unit tests, which confirms the correctness of the implementation rather than the benefit of the method. It has not been run end to end, and no results are reported for it.

Probe, score, consensus, track. Within each window every geometrically plausible camera is probed on a single frame, each view is scored by visible cross-section in square metres rather than in pixels for the reasons developed in Section 4.2.4, views whose observations are geometrically inconsistent with the others are rejected by consensus, and the surviving best three are tracked through the window. Tracking three views rather than one is not a robustness measure.

This replaces an earlier design that tracked a single best camera per frame, selected by depth and stabilised with hysteresis. That design cannot serve reconstruction, which requires simultaneous visibility in at least three views.

Crops and prompts. Crops follow the hand-derived 3D position projected into each selected view. Because that position is offset by the grip distance and degrades as Table 8 describes, crops are sized to contain generously rather than to centre precisely. Crops sizes are visible in Figure 5. Prompts are the base text for each object taken from `object_prompts.json`, with one SAM 3 session per object per view and no joint prompting across objects. The choice of base rather than enriched prompts is not a default but a result, reported in Section 4.3.3.

4.3 Experiments and Results

4.3.1 Benchmark Evaluation Across the Three Conditions

Table 11 reports the per-clip scores. One correction has been applied relative to the figures circulated in the bi-weekly report of 23 July and in the accompanying presentation, and it is stated here rather than buried because it reverses a conclusion. The clips containing mushrooms were originally scored at 5 percent with the observation that only one instance was segmented. That observation was an artefact of the visualisation, which drew only the highest-scoring masklet. Re-examination of the persisted run-length encodings shows approximately fourteen mushrooms segmented with distinct identities and a mean pairwise intersection over union of 0.00, which is to say genuinely distinct instances rather than duplicates of one. The inference drawn at the time, that multi-instance segmentation is a weak point of SAM 3, does not survive the correction.

Clip	Score	Observation
<i>Condition 1: present, not handled</i>		
Cutting board, carrots cut above it	100%	Detection governed by illumination; recovers once the subject shades the blade ≈ 14 instances, distinct ids, mean pairwise IoU 0.00
Cutting board, mushrooms cut above it	100%	
Knife, unoccluded	100%	
Knife, tilted, specular	0% $\rightarrow$ 50%	
Mushrooms	0%	
Carrot packaging	100%	
Pasta packaging	100%	
Pan at frame edge	100%	
Spatula in pan, heavily occluded	100%	
Spatula in pot of water	100%	
<i>Condition 2: handled, identity preserved</i>		
Doser pouring water into a pan	100%	Segments the zucchini on one instance; no mask on three frames, plausibly blade angle
Knife cutting zucchini	75%	
Mushrooms	0%	
Carrot package	0%	Out of frame, the subject's head is segmented instead
Pasta	0%	
Spatula stirring	100%	
Spatula stirring at frame edge	100%	Correct in frame; out of frame, the knife or part of a plate is returned
Spoon in the left hand	50%	
Towel drying hands	100%	
<i>Condition 3: transformed by the action</i>		
Bell pepper washed, emptied, cut	60%	Washing and handling 100%; halved 100%; emptied 50%, one of two masks lost; cut and re-washed 0%
Onion cut	50%	Fragments not segmented
Onion cut	50%	Fragments not segmented
Zucchini cut	50%	Only the fragments are segmented

Table 11: SAM 3 benchmark across the three conditions. The score is only a visual assessment of how well SAM 3 segments the given object. Handling does not degrade performance between Conditions 1 and 2; degradation appears only at Condition 3, and even there it is partial and object-dependent.

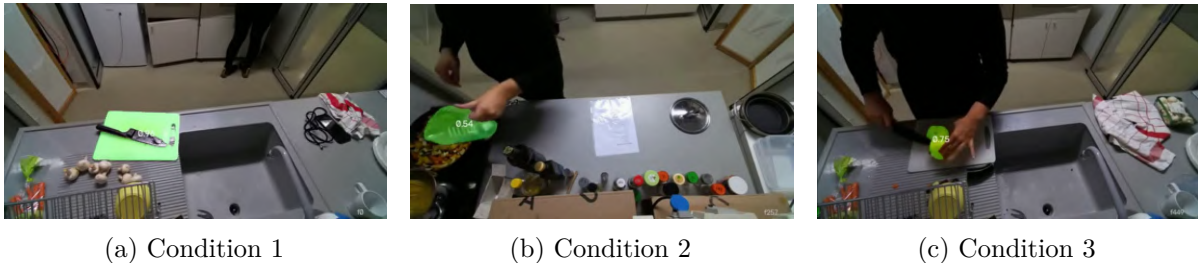


Figure 9: Multi-view observations across three synchronized camera perspectives.

The comparison that matters is between Conditions 1 and 2. Handling does not degrade performance. Objects that are carried, stirred with, poured from and wiped with are segmented as reliably as objects sitting on a counter; the spatula is tracked through heavy intermittent occlusion, and the cutting boards are fully occluded by a person working above them and score

perfectly. The working hypothesis that motivated the interaction module, that manipulation and occlusion are the dominant failure modes, is not supported. Degradation appears at Condition 3, where the object ceases to be the thing the prompt names.

4.3.2 Failure Mode Taxonomy

The failures observed share a property more consequential than their frequency. With the qualification noted at the end of this subsection, they are false positives or identity errors rather than absences. The model rarely declines to answer; it answers with something else. Table 12 sets out the taxonomy and what each mode implies for the design.

Mode	Description and consequence
Substitution under absence	The prompted object leaves the field of view and a look-alike is returned in its place: the subject’s head for a package, a knife or part of a plate for a spoon. The most damaging mode, because the consumer receives confident wrong data rather than missing data.
View-dependent collapse	Glare and blade tilt reduce detection to nothing in one view while other views are unaffected. The failure is a property of the viewpoint, not of the object or the interaction, which is what makes another camera a sufficient remedy.
Extension loss under transformation	After a state change the object no longer falls under the noun phrase that retrieved it, so the concept stops matching. Not a tracking failure.

Table 12: Failure taxonomy. None of the four modes is specific to hand–object interaction, and three of the four are resolvable from calibrated geometry to which the segmenter has no access.

The practical significance is that this taxonomy forecloses the obvious remedy. If failures were absences, a confidence threshold would separate the reliable outputs from the unreliable ones. Because they are identity errors, a hallucinated mask carries a score comparable to a correct one, and no threshold separates them. Correction must therefore come from a signal the segmenter does not have, which in this setting is the calibrated geometry of the rig. This is the argument for the orchestration design of Section 4.2.3, and it is an argument from the structure of the errors rather than from their rate.

The same property invalidates the metric that would otherwise be natural here. The proportion of frames on which a mask is returned, the empty rate, is broken in both directions at once: an object genuinely absent and correctly reported as such, as with the Trash object at a 100 percent empty rate, scores worst, while a confident hallucination scores well. We must score identity rather than presence.

This taxonomy also bears on a published prediction. TubeletGraph [32] reports that SAM 2 achieves precision above 70 percent with recall below 55 percent on transforming objects, a profile in which false negatives dominate, and anticipates that text-prompted models will behave similarly. The observations above are the opposite: for SAM 3 on this data false negatives are rare and false positives are characteristic. The divergence is explicable from the architectures described in Section 2.2, since SAM 2 propagates masks forward through a memory bank and loses objects it cannot follow, whereas SAM 3 re-detects from text at every frame and consequently always has a candidate to offer. The observation is worth reporting and should be reported with its limits: one session, one rig, one object set, and a qualitative score.

4.3.3 Prompt Specificity and the Extension of a Concept

The hypothesis that action annotations could be injected into segmentation through the prompt itself was tested directly, comparing base prompts naming only the object against prompts enriched with the annotated verb, as in `mushrooms being cut`. Enrichment was expected to supply temporal and spatial context at no architectural cost. It degrades performance instead, and the mechanism is worth stating precisely because it is not a matter of prompt engineering quality.

Adding the verb narrows the extension of the concept. Writing $E(\cdot)$ for the set of objects a phrase denotes,

$$E(\text{"mushrooms being cut"}) \subsetneq E(\text{"mushrooms"}),$$

the rest of the set is the uncut mushrooms. Since the annotations are recorded per noun over an interval, the enriched prompt is simply false for part of that interval, and the segmenter behaves correctly by declining to return objects that do not match the description it was given. The failure is one of specification, not of the model.

Two further obstacles surfaced during the same experiments, and both are worth recording because they shaped what was possible afterwards. The first was mechanical. In the initial implementation the prompt was recomputed at every change of annotated phase, and changing it restarted the SAM 3 video session, which destroyed the memory bank precisely at the moments the enrichment was supposed to help. The intent of enriching a prompt is to add temporal information, so an implementation that discards temporal state in order to apply it is self-defeating.

The resolution is to resolve every prompt the window will require before inference begins, rather than one phase at a time. For a single event of the working session this amounts to a short indexed list:

```
all_prompts = [
    "yellow bell pepper",           # 0   no action annotated
    "yellow bell pepper being carried", # 1   Carry
    "yellow bell pepper being cut",   # 2   Cut
]
```

All three are registered once, and the frame-by-frame decision becomes a choice of index into that list instead of a reinstantiation. Prompt changes therefore no longer cost the memory bank. The only session restarts that remain are those imposed by the memory-bank cap discussed in Section 4.2.3, and this separation is what later made a multi-prompt strategy conceivable at all. The second obstacle was economic, and it turned out to be milder than assumed. Testing several prompts was initially treated as multiplying the number of jobs, which would have made any multi-prompt scheme unaffordable at the scale of EPFL-Smart-Kitchen-30. Inspection of the model shows otherwise: the vision backbone runs once per frame and its output is independent of the prompt, at roughly 850 ms per frame, while detection is a sequential loop over prompts costing roughly 210 ms each. The cost of N prompts on a frame is therefore $850 + 210N$ milliseconds rather than N times the single-prompt cost, so three prompts cost about 40 percent more than one instead of three times as much.

That margin made it practical to ask where the concept actually dies. Six prompts of varying specificity were evaluated on a single frame of the working session, after the bell pepper had been diced, each in its own single-frame session so that no temporal memory could carry information between them and only the text differed. Figure 10 shows the result. Prompts naming the object precisely recover little or nothing of the pile, and the base prompt used by the pipeline fails outright, while the coarsest description, which drops the specific noun in favour of a category term, recovers the pile almost in full.

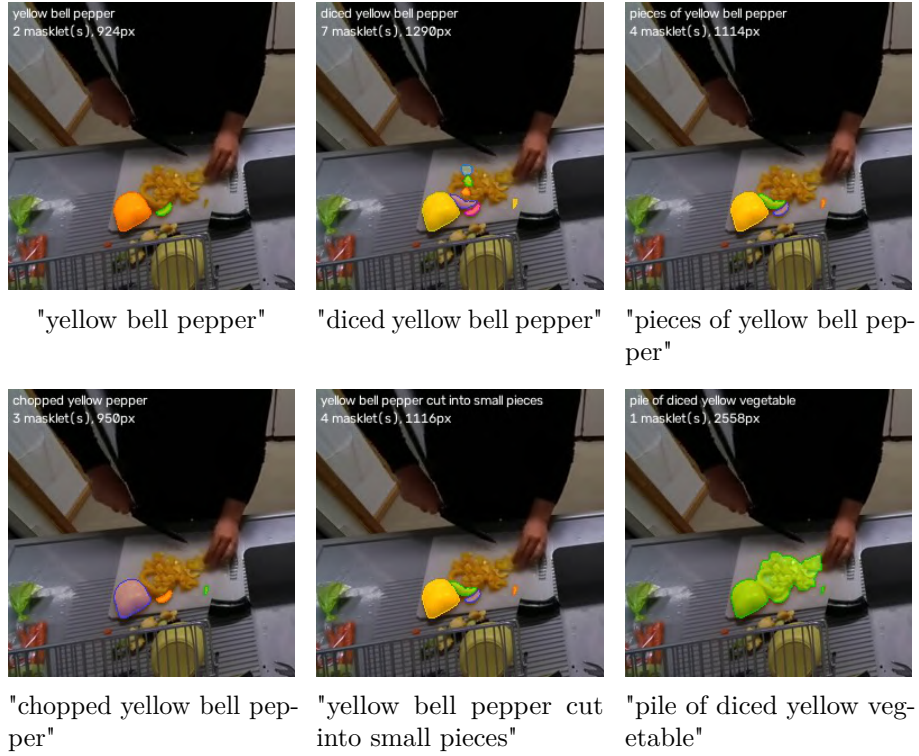


Figure 10: Six prompts of decreasing specificity applied to the same frame after the bell pepper has been diced, each in an independent single-frame session so that only the text differs.

We can clearly see a compromise between specificity and extension. Such a module could be implemented but the integration in SAM 3's architecture is not straightforward. The model is not designed to handle multiple prompts in a single session, and the current implementation would require significant modifications to accommodate this functionality.

4.4 Cost and the Quantified Form of the Orchestration Argument

Inference costs 1.06 seconds per frame per object on a V100 for a single prompt, measured over the pipeline's own execution. The internal structure of that cost determines what can be optimised. As established in Section 4.3.3, the vision backbone runs once per frame and its output is reused both by the detection stage, which loops over the registered prompts, and by the tracker. The prompt-independent majority of the cost is therefore shared across all objects present in a frame, and the cost of an additional object is negligible.

Figure 11 gives the resulting budget for one full session.

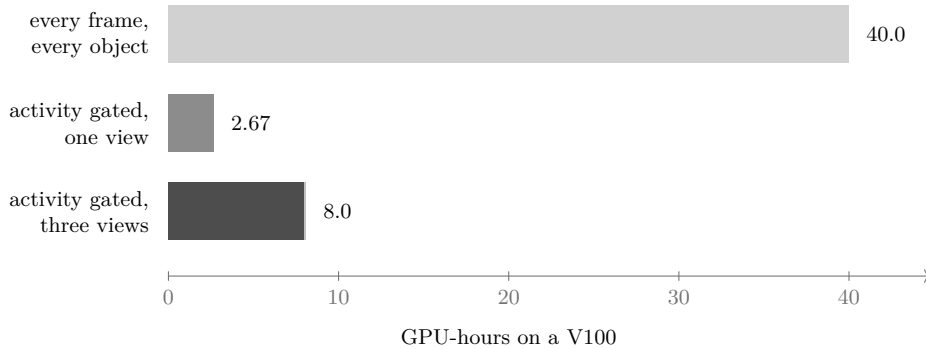


Figure 11: GPU budget for one full session.

Processed without gating, a full session would require approximately 40 GPU-hours. Restricting processing to the intervals in which an object is under an annotated action, which covers between five and seven percent of the timeline per object, reduces this to 2.67. Tracking three views rather than one returns it to roughly eight. Against those figures the probe is negligible, so the objection that segmenting every candidate camera would be prohibitively expensive does not survive measurement: what is expensive is tracking, and the annotations control how much of it is done.

4.5 Limitations

The principal weakness of this work is evidential rather than conceptual. The benchmark is qualitative, its sample is small and selected by me, Condition 3 rests on four clips, and all of it comes from a single session on a single rig. The conclusions drawn concern the direction and structure of failures rather than their rates, which is what a sample of this kind supports, but the step from here to a defensible quantitative claim requires ground truth. The annotation infrastructure for producing it, described in Section 4.2.1, was built for a different purpose and remains available, which makes closing this gap the natural next step rather than a distant one. Moreover, some leads are still to be explored. The most notable one being the multi-view SAM 3 latent space. Indeed there is no refactored architecture of SAM 3 that allows for multi-view segmentation. It could be a promising direction to have a module that predicts the camera in which the object lives and be able to project segmentations through other cameras.

5 Discussion

The two projects were posed independently and solved different problems, but they converged on the same shape of answer, and the convergence is worth stating because it was not planned.

Neither contribution is a model. STroMAE does not propose a new architecture: it initialises from a released VideoMAE trunk and leaves masked pretraining untouched, and what it contributes is the layer that decides which embedders exist, which heads are attached and how a dataset is resolved, so that three diverging forks become one declarative surface. The segmentation pipeline does not propose a new segmenter: SAM 3 is frozen throughout, and what the pipeline contributes is the layer that decides when the model is worth running, which views are worth trusting, and which of its proposals to keep. In both cases the useful work sat around a fixed model rather than inside it, and in both cases the signal that made it possible was one the model had no access to: a declared experiment specification in the first, and the calibration and action annotations of the rig in the second.

For the segmentation pipeline this was a correction rather than a starting point. The work began from the hypothesis that SAM 3 would struggle under hand-object occlusion, and that a module conditioned on hand pose and action context could help it see. The benchmark of Section 4.3.1 falsified that premise. Carrying, stirring and pouring are handled, multiple instances of a noun are separated correctly, and fragments produced by cutting continue to receive masks. What fails is identity: when the target leaves the frame a visually similar object takes its place, and the subject’s own head is returned as an onion or a package. This matters beyond the taxonomy, because a confidently wrong mask scores no differently from a correct one, so no confidence threshold can repair it. The project therefore moved from learning an interaction prior to orchestrating a frozen segmenter, which is how it arrived at the same posture as STroMAE by a different route.

The second theme the two projects share is less flattering and more useful. In both, the hardest defects were the ones that produced plausible numbers instead of errors. In STroMAE the inherited training loop selected its loss branch by the head’s metric rather than by its declared loss, so a head could be optimised through the wrong objective while training proceeded

normally; the defect surfaced only because the loss settled on 0.5623, which is exactly $H(0.25)$, the binary entropy of that configuration’s marginal label rate. In the segmentation pipeline the memory-bank resets were judged harmless because the number of masklets held steady across them, and only visual inspection established that the entire pile of diced pepper had been lost: a count can hold constant, or improve, while segmented area collapses, since losing one large mask and gaining two spurious small ones looks like progress. An earlier rendering defect had drawn only the highest-scoring masklet, which for some time made a correctly segmented tray of mushrooms look like a failure to separate instances.

The common lesson is that both projects were limited less by what the models could do than by whether the measurements were actually meaningful. The overfitting entry criterion in Section 3.2.5 and the switch from masklet count to mask area are the same methodological move made in two settings: refuse to interpret any full-scale number until a diagnostic exists that a degenerate solution cannot pass.

The prompt results of Section 4.3.3 deserve a separate remark, because they are easily read as one negative and one positive finding when they are the same finding seen twice. Enrichment fails because it makes the description too specific to remain true of the object. Transformation fails because it moves the object out of a description that was true until then. Both are subject to a compromise, so no single fixed choice is correct for a whole timeline.

Several limitations should be stated plainly. On the STrOMAE side, only the video embedder inherits pretrained weights, while the audio, segmentation, pose and auxiliary-view embedders begin from random initialisation and acquire meaning only during fine-tuning, which is the asymmetry the design accepts in exchange for not requiring multimodal pretraining, and it is likely to bound what the non-video streams can contribute. The deviations from the published protocols listed in Section 2.1 mean the comparisons should be read as situating the framework rather than as like-for-like benchmark results. On the segmentation side, the only per-frame three-dimensional signal available for an object is derived from the hand presumed to hold it, so every geometric test is strictly a test of proximity to the hand, and it is weakest precisely during cutting, when the pieces rest on the board and the hand is on the knife. The calibration is assumed sound rather than verified, no reprojection check was performed, and frame-level synchronisation between cameras is likewise assumed and never tested, although a one-frame offset at thirty frames per second would introduce several centimetres of triangulation error without any visible symptom. Finally, the multi-view configuration of Section 4.2.5 is implemented and unit-tested but has not been run end to end, so the reconstruction objective that motivated it remains unevaluated.

6 Conclusion and Future Work

This report has described two contributions made at the Mathis Group between May and September 2026, addressing the two recurring obstacles identified in Section 1: scaling one representation model across arbitrary input modalities, and extracting pixel-level object information from a multi-camera rig.

STrOMAE answers the first. A single architecture and a single training loop now serve any combination of modalities and any set of prediction heads, with each stream mapped into a shared token space by its own embedder and fused by full self-attention in a shared trunk. An experiment is specified declaratively, so adding a dataset requires configuration rather than a new loader or a new training script, and the three diverging forks the laboratory maintained are replaced by one surface. The framework reproduces the reference benchmarks on both EPFL-Smart-Kitchen-30 and MammAlps, and the controlled overfitting protocol adopted before any full run surfaced several defects in the inherited codebase that had never raised an error. The segmentation pipeline answers the second, though not in the form the question was posed. Its first result is a characterisation of how SAM 3 fails on EPFL-Smart-Kitchen-30 footage that

contradicts the premise the project started from, and its second is a pipeline that produces per-object masks over a full session by orchestrating that frozen model with the signals the dataset provides: the action annotations bound when an object is under manipulation, and the calibration bounds where it can be seen. Activity gating alone reduces the cost of a full session from approximately forty GPU-hours to 2.67. Alongside these, the work produced a negative result on prompt enrichment stated precisely enough to be reused rather than merely reported. STrOMAE is being released publicly through the laboratory’s repository at <https://github.com/amathislab/> and packaged for distribution on PyPI, so that it can be installed and reused directly rather than forked.

The segmentation work is being prepared for submission to CVPR. The schedule is tight, and what determines feasibility is not the pipeline but the quantitative evaluation it currently lacks. The multi-view configuration has not been run end to end, and no manually annotated ground truth yet exists against which the produced masks can be scored. Both are prerequisites rather than extensions, and both are addressed below.

The immediate work is therefore evaluation rather than design. A set of manually annotated frames, drawn from the failure modes already catalogued rather than sampled uniformly, would turn qualitative observations into measurements. All masks are stored as run-length encodings precisely so that runs already completed can be scored retroactively once such annotations exist. The multi-view configuration needs to be run over a full session, and the two assumptions it rests on, calibration accuracy and inter-camera synchronisation, should be checked first, since both are cheap to test and either would invalidate everything downstream if it failed.

Three improvements follow directly from results already obtained. Aligning memory-bank resets with the annotations, so that the session is reset immediately before a transforming action rather than at a fixed frame count, would let the tracker carry through a transformation instead of being forced to re-detect, in the middle of one, a concept that no longer matches. Selecting the prompt during execution, by evaluating a short ladder of candidate descriptions at the moment the probe already runs, would address the specificity problem without committing to one description for an entire timeline. And the rate of decay of mask area, which is computed but currently unused, is an early warning that a view has become unreliable several tens of frames before the geometric criteria notice.

That objective is what connects this work to its clinical motivation. Recovering where the manipulated objects are, and not only what action was performed, is what would allow gaze to be related to the objects a person is actually reaching for, which is the measurement the stroke rehabilitation application requires and the one EPFL-Smart-Kitchen-30 was instrumented to make possible. The contribution of this project is a route to that measurement which does not depend on dense ground-truth segmentation masks, and therefore does not depend on annotating by hand what the rig can be made to infer.

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